

CHAPTER 1

Magnetic Neutron Scattering

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Contents

1. Introduction	3
2. Basic properties of the neutron	3
3. Neutron source	5
4. Neutron scattering	5
4.1. Definitions of scattering cross-section	6
4.2. The master equation	7
5. Nuclear neutron scattering	9
5.1. Neutron scattering from a single nucleus	9
5.2. Coherent and incoherent scattering	11
6. Magnetic neutron scattering	12
6.1. Scattering of neutrons from unpaired electrons	12
6.2. Scattering of neutrons from crystalline magnetic materials	15
6.3. Elastic magnetic scattering from crystals	15
6.4. Inelastic magnetic scattering	17
6.5. Spin waves	18
6.6. Scattering from spin waves	20
6.7. Paramagnetic scattering	21
6.8. Crystal-field excitations	22
7. Concluding remarks	23
References	23

NEUTRON SCATTERING FROM MAGNETIC MATERIALS

Edited by Tapan Chatterji

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1. Introduction

The discovery of the neutron in 1932 by Chadwick [1] certainly had the most profound consequences. The era of nuclear physics began culminating into nuclear technology and the particle physics was born. Elsasser [2] was the first to suggest that the motion of neutrons would be determined by wave mechanics and thus would be diffracted by crystalline materials. The first demonstration of the diffraction of neutrons was done by Halban and Preiswerk [3] and also by Mitchell and Powers [4]. These experiments were done using a radium–beryllium neutron source. The scattering process was of nuclear origin, i.e., the neutrons were scattered by nuclei. The idea of magnetic neutron scattering originated from Bloch [5]. He wrote a two-page letter to the editor of the *Physical Review* in which he suggested that if the value of the magnetic moment of the neutron was of the same order as the known measured magnetic moment of the proton, then neutron scattering by the spin and orbital moments of magnetic atoms should be observable. Later Alvarez and Bloch [6] showed experimentally that the neutron magnetic moment was about 0.7 of the proton value. A detailed discussion of the magnitude of magnetic neutron scattering was given by Halpern and Johnson [7]. Following the prediction of antiferromagnetism by Néel [8], Shull and Smart [9] provided the first experimental evidence of this phenomenon in MnO by neutron diffraction. Starting with these developments, decades of research in magnetic neutron scattering followed and is still contributing enormously to the microscopic understanding of condensed matter. The principles of magnetic neutron scattering have been treated in several excellent books [10–17] and readers are advised to consult them. Here we attempt to summarize the essential aspects of magnetic neutron scattering.

2. Basic properties of the neutron

The scattering of slow neutrons is a very powerful technique to investigate the structure and dynamics of condensed matter. The usefulness of this technique stems from the fundamental properties of the neutron summarized in Table 1.

The value of the mass of the neutron $1.674928(1) \times 10^{-24}$ g leads to a de Broglie wavelength of thermal neutrons of about 1.8 Å which is of the order of the interatomic distances in condensed matter making interference effects possible. Thus neutron scattering can yield structural information about condensed matter. The energies of thermal and cold

Table 1
Basic properties of the neutron

Property	Value or equation
Mass	$1.674928(1) \times 10^{-24}$ g
Charge	0
β -decay life time	885.8 ± 0.9 s
Free neutron decay	$n \rightarrow p + e^- + \bar{\nu}_e$
Spin	$\frac{1}{2}$
Magnetic moment, μ_n	$-1.9130427(5)\mu_N$

neutrons being in the range from 0.1 meV to 100 meV match very well with energies of elementary excitations of condensed matter. This enables one to investigate the dynamics of the elementary excitations of condensed matter. For excitations with energies higher than 100 meV hot neutrons can also be used. The property that the neutron is a chargeless particle enables the neutron to penetrate deep into condensed matter making bulk investigation feasible. Also, apparatus for extreme sample environments can be fabricated relatively easily. The neutron has a magnetic moment and therefore it can interact with unpaired electrons of magnetic atoms. Thus neutron scattering can yield valuable information about the magnetic structure and spin dynamics of magnetic materials. The neutrons obtained from neutron sources are usually unpolarized. However, the neutrons can be polarized relatively easily. Polarized neutron scattering can yield more detailed and unambiguous information about the magnetic structure, magnetization distribution and spin dynamics of condensed matter.

For the purpose of neutron scattering investigations there exist two types of neutron sources, viz. reactor and spallation neutron source. The neutrons emerging from these sources have very high energies (epithermal neutrons) and are therefore moderated to have useful energy ranges. The neutrons are called thermal, cold or hot depending on the temperature T of the moderator. The probability of neutrons having a velocity between v and $v + dv$ follows closely the Maxwell–Boltzmann probability distribution,

$$P(v) dv = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 \exp\left(-\frac{1}{2} \frac{mv^2}{k_B T}\right) dv, \quad (1)$$

where m is the mass of the neutron and k_B is the Boltzmann constant. The maximum of the function $P(v)$ occurs at a velocity v that corresponds to the kinetic energy of the neutron E which is given by

$$E = \frac{1}{2}mv^2 = k_B T. \quad (2)$$

The de Broglie wavelength λ and the magnitude of the wave vector k of the neutron can be expressed as

$$\lambda = \frac{h}{mv}, \quad (3)$$

$$k = \frac{2\pi}{\lambda}. \quad (4)$$

The momentum $\mathbf{p}$ and the wave vector $\mathbf{k}$ of the neutron are related by the equation

$$\mathbf{p} = \hbar \mathbf{k}. \quad (5)$$

The energy E of the neutron can be related to the temperature T , velocity v , momentum p , magnitude of wave vector k , wavelength λ and the frequencies ω and ν ($\omega = 2\pi\nu$) by

$$E = k_B T = \frac{1}{2}mv^2 = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m} = \frac{h^2}{2m\lambda^2} = \hbar\omega = h\nu. \quad (6)$$

Table 2
Values of some physical constants

Physical constant	Value
Planck constant, h	$6.6260755(40) \times 10^{-34} \text{ J s}$
Boltzmann constant, k_B	$1.380658(12) \times 10^{-23} \text{ J K}^{-1}$
Bohr magneton, μ_B	$9.2740154(31) \times 10^{-24} \text{ J T}^{-1}$
Nuclear magneton, μ_N	$5.0507866(17) \times 10^{-27} \text{ J T}^{-1}$

Taking values of the physical constants from Table 2 we get the following useful equations:

$$E = 0.08617T = 5.227v^2 = 2.072k^2 = 81.81 \frac{1}{\lambda^2}$$

$$= 0.685 \times 10^{-12} \omega = 4.136\nu, \quad (7)$$

$$\lambda = 6.283 \frac{1}{k} = 3.956 \frac{1}{v} = 9.045 \frac{1}{\sqrt{E}} = 30.81 \frac{1}{\sqrt{T}}, \quad (8)$$

where λ is in Å, k is in Å⁻¹, v is in km s⁻¹, E is in meV, T is in K, ω is in s⁻¹ and ν is in THz.

3. Neutron source

We already mentioned that there exist two main types of neutron sources at which the neutron scattering experiments can be done, namely the reactor and spallation neutron sources. Most of the experiments discussed in this book were done at reactor based neutron sources. Among the existing reactor based neutron sources, the 57 MW high flux reactor at the Institut Laue–Langevin in Grenoble is the most powerful and has contributed much to the neutron scattering investigation of condensed matter in general and of the magnetic properties of condensed matter in particular.

4. Neutron scattering

Neutrons are scattered by the nuclei and also by the unpaired electrons of the magnetic atoms in condensed matter. The corresponding neutron scattering is called nuclear neutron scattering and magnetic neutron scattering, respectively. In the present book we will mainly consider magnetic neutron scattering. However, neutron scattering intensity from magnetic materials is a superposition of both types of scattering. In order to be able to separate magnetic scattering from nuclear scattering and to extract information about the magnetic structure and spin dynamics, it is important to understand the basic principles of both processes. In the present section we will give definitions and describe some basic principles of neutron scattering in general which are valid both for nuclear and magnetic neutron scattering. In the following sections we will describe nuclear neutron scattering briefly followed by a more detailed treatment of magnetic neutron scattering.

4.1. Definitions of scattering cross-section

The double differential scattering cross-section is defined by the equation

$$\frac{d^2\sigma}{d\Omega dE} = \frac{C}{\eta\Phi N\Delta\Omega\Delta E}, \quad (9)$$

where Φ is the flux of incident neutrons, i.e., the number through unit area per second, the area being perpendicular to the direction of the neutron beam, C is the count rate for a given interval ΔE of energy transfer in the detector which subtends a solid angle of $\Delta\Omega$ and has an efficiency η and N is the number of identical atoms in the beam. The number of identical scatterers N is introduced such that the scattering cross-sections are expressed per scatterer (per atom). In subsequent formulae for the neutron scattering cross-section we have divided the right-hand side by the number N or N_m (the subscript “m” stands for “magnetic”) so that the scattering cross-section is expressed per atom or per magnetic atom. This number depends on the summation carried out on the right-hand side (total number of scatterers, total number of magnetic atoms or total number of magnetic atoms per unit cell).

If we do not analyze the energy but simply count all the neutrons scattered into a solid angle $d\Omega$ in the direction θ, ϕ , then the corresponding cross-section, known as the differential cross-section, is defined by the equation (see Figure 1)

$$\frac{d\sigma}{d\Omega} = \frac{C}{\eta\Phi N\Delta\Omega}. \quad (10)$$

If we count the total number of neutrons scattered in all directions we can define a total scattering cross-section by the equation

$$\sigma_{\text{tot}} = \frac{C}{\eta\Phi N}. \quad (11)$$

From their definitions the three cross-sections are related by the following equations

$$\frac{d\sigma}{d\Omega} = \int_0^\infty \left(\frac{d^2\sigma}{d\Omega dE'} \right) dE', \quad (12)$$

$$\sigma_{\text{tot}} = \int_{\text{all directions}} \frac{d\sigma}{d\Omega} d\Omega. \quad (13)$$

If the scattering is axially symmetric, i.e., the scattering depends only on θ and not on ϕ , the above equation becomes

$$\sigma_{\text{tot}} = 2\pi \int_0^\pi \frac{d\sigma}{d\Omega} \sin\theta d\theta. \quad (14)$$

The scattering of neutrons by condensed matter from an incoming state characterized by a wave vector $\mathbf{k}_0$ and a spin σ_0 into an outgoing state characterized by a wave vector $\mathbf{k}_1$

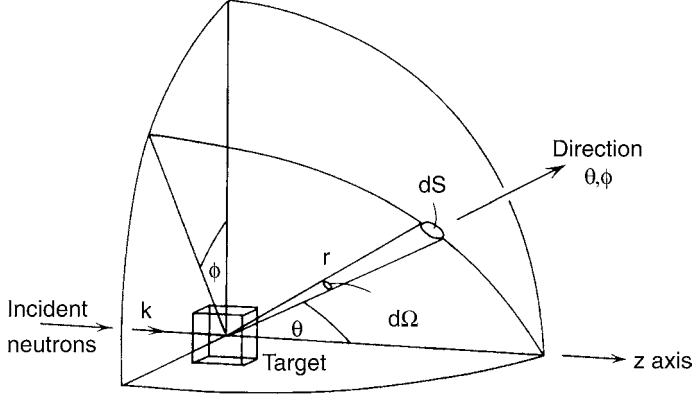


Fig. 1. Geometry for scattering experiment (after Squires [12]).

and a spin σ_1 can be represented by the differential scattering cross-section $d\sigma/d\Omega$. In a neutron scattering experiment the count rate C in a detector that makes a solid angle $d\Omega$ and has an efficiency η is given by

$$C = N\Phi d\Omega \eta \left(\frac{d\sigma}{d\Omega} \right)_{\mathbf{k}_0, \sigma_0 \rightarrow \mathbf{k}_1, \sigma_1}. \quad (15)$$

4.2. The master equation

The theory of neutron scattering has been treated in several textbooks [10,12–14,16] and articles [15,18]. We recall here some basic results. We consider scattering of neutrons by a sample consisting of condensed matter which undergoes a change from a state λ_0 to a state λ_1 while the state of the neutron changes from $(\mathbf{k}_0, \sigma_0)$ to $(\mathbf{k}_1, \sigma_1)$. The corresponding differential scattering cross-section is given by

$$\left(\frac{d\sigma}{d\Omega} \right)_{\mathbf{k}_0, \sigma_0, \lambda_0 \rightarrow \mathbf{k}_1, \sigma_1, \lambda_1} = \frac{1}{N\Phi d\Omega} \sum_{\mathbf{k}_1} W_{\mathbf{k}_0, \sigma_0, \lambda_0 \rightarrow \mathbf{k}_1, \sigma_1, \lambda_1}, \quad (16)$$

where $W_{\mathbf{k}_0, \sigma_0, \lambda_0 \rightarrow \mathbf{k}_1, \sigma_1, \lambda_1}$ is the number of transitions per second from the state $\mathbf{k}_0, \sigma_0, \lambda_0$ to the state $\mathbf{k}_1, \sigma_1, \lambda_1$ and Φ is the flux of incident neutrons. The summation is over all values of $\mathbf{k}_1$ that lie in the small solid angle $d\Omega$ in the direction θ, ϕ , the values k_0, λ_0 and λ_1 remaining constant. The right-hand side of the above equation is evaluated by using Fermi's golden rule,

$$W_{\mathbf{k}_0, \sigma_0, \lambda_0 \rightarrow \mathbf{k}_1, \sigma_1, \lambda_1} = \frac{2\pi}{\hbar} |\langle \mathbf{k}_1 \sigma_1 \lambda_1 | V | \mathbf{k}_0 \sigma_0 \lambda_0 \rangle|^2 \rho_{\mathbf{k}_1 \sigma_1}(E_1), \quad (17)$$

where V is the interaction potential between the neutron and the sample and $\rho_{\mathbf{k}_1 \sigma_1}(E_1)$ is the density of the final scattering states per unit energy interval. The golden rule is based

on the validity of first-order perturbation theory. It is certainly valid for nuclear neutron scattering since the nuclear potential is short range and only s -wave scattering is possible. The magnetic scattering potential is not short range but it is weak and therefore the golden rule is still valid. To calculate the matrix element in (17) we consider the neutron and the sample in a large box of volume V_0 , the incident and scattered neutron functions being $V_0^{-1/2}e^{i\mathbf{k}_0 \cdot \mathbf{r}}|\sigma_0\rangle$ and $V_0^{-1/2}e^{i\mathbf{k}_1 \cdot \mathbf{r}}|\sigma_1\rangle$, respectively. The number of states of the scattered neutron in the energy interval dE_1 is

$$\rho_{\mathbf{k}_1\sigma_1(E_1)} = \frac{V_0}{8\pi^3} d\mathbf{k}_1 = \frac{V_0}{8\pi^3} k_1^2 dk_1 \Delta\Omega.$$

Using the known result of quantum mechanics

$$dE_1 = \frac{\hbar^2 k_1 dk_1}{m}$$

we have

$$\rho_{\mathbf{k}_1\sigma_1(E_1)} = \frac{V_0}{8\pi^3} \frac{mk_1}{\hbar^2} \Delta\Omega. \quad (18)$$

The incident neutron flux is equal to the normalized number density times the neutron velocity

$$\Phi = \frac{v}{V_0} = \frac{\hbar k_0}{V_0 m}. \quad (19)$$

Substituting equations (17)–(19) in (16), we get

$$\left(\frac{d\sigma}{d\Omega} \right)_{\mathbf{k}_0, \sigma_0, \lambda_0 \rightarrow \mathbf{k}_1, \sigma_1 \lambda_1} = \frac{1}{N} \frac{k_1}{k_0} \left(\frac{m}{2\pi\hbar^2} \right)^2 |\langle \mathbf{k}_1 \sigma_1 \lambda_1 | V | \mathbf{k}_0 \sigma_0 \lambda_0 \rangle|^2, \quad (20)$$

where $|\mathbf{k}_0\rangle$, $|\mathbf{k}_1\rangle$ denote the plane waves $V_0^{-1/2}e^{i\mathbf{k}_0 \cdot \mathbf{r}}$, $V_0^{-1/2}e^{i\mathbf{k}_1 \cdot \mathbf{r}}$. From the law of conservation of energy

$$E = E_0 - E_1 = E_{\lambda_1} - E_{\lambda_0}, \quad (21)$$

we get

$$\begin{aligned} & \left(\frac{d^2\sigma}{d\Omega dE} \right)_{\mathbf{k}_0, \sigma_0, \lambda_0 \rightarrow \mathbf{k}_1, \sigma_1 \lambda_1} \\ &= \frac{1}{N} \frac{k_1}{k_0} \left(\frac{m}{2\pi\hbar^2} \right)^2 |\langle \mathbf{k}_1 \sigma_1 \lambda_1 | V | \mathbf{k}_0 \sigma_0 \lambda_0 \rangle|^2 \delta(E + E_{\lambda_0} - E_{\lambda_1}). \end{aligned} \quad (22)$$

We now sum over all final states of the sample λ_1 and final polarization states σ_1 , average over all initial states λ_0 of the sample, which occur with probability p_{λ_0} , and over all initial states of the neutron, which occur with a probability p_{σ_0} to obtain

$$\begin{aligned} & \left(\frac{d^2\sigma}{d\Omega dE} \right)_{\mathbf{k}_0 \rightarrow \mathbf{k}_1} \\ &= \frac{1}{N} \frac{k_1}{k_0} \left(\frac{m}{2\pi\hbar^2} \right)^2 \sum_{\lambda_0\sigma_0} p_{\lambda_0} p_{\sigma_0} \sum_{\lambda_1\sigma_1} |\langle \mathbf{k}_1\sigma_1\lambda_1 | V | \mathbf{k}_0\sigma_0\lambda_0 \rangle|^2 \delta(E + E_{\lambda_0} - E_{\lambda_1}). \end{aligned} \quad (23)$$

Equation (23) is called the master formula of neutron scattering which forms the basis for the interpretation of all experimental neutron scattering data. This is a very general equation in which we make no assumption about the interaction potential $V(\mathbf{r})$ between the sample and the neutron.

5. Nuclear neutron scattering

5.1. Neutron scattering from a single nucleus

The scattering of neutrons by a nucleus is caused by the nuclear forces which have a range of about 10^{-12} – 10^{-13} cm. The wavelength of the thermal neutron is of the order of 10^{-8} cm which is much larger than the range of nuclear forces. It is well known from the theory of scattering that the scattered wave in such cases is spherically symmetric, i.e., analyzed in terms of partial waves, has the character of an s wave ($l = 0$). We take the origin to be at the nucleus and the z axis to be along the neutron wave vector $\mathbf{k}$. If the incident neutron is represented by the wave function

$$\psi_{\text{inc}} = \exp(ikz), \quad (24)$$

then the wave function of the scattered wave at the point $\mathbf{r}$ is given by

$$\psi_{\text{sc}} = -\frac{b}{r} \exp(ikr), \quad (25)$$

where b is a constant independent of the angles θ and ϕ . The minus sign in the equation is arbitrary and corresponds to a positive value of b for a repulsive potential. For simplicity we have considered the case of elastic scattering (fixed nucleus) and therefore the wave vector of the neutron k remains the same for the scattered wave. The quantity b which has a dimension of length is defined as the scattering length. The scattering length b is in general a complex quantity given by

$$b = b_0 + b' + ib'', \quad (26)$$

where b_0 is the length associated with the potential scattering and b' and ib'' are the real and imaginary parts of the resonance scattering which takes place with formation of the compound nucleus. The imaginary term signifies absorption of neutrons by the nucleus and is important for a few strongly absorbing nuclei like ^{103}Rh , ^{113}Cd , ^{157}Gd , ^{176}Lu , etc. The scattering length for such nuclei is strongly dependent on the incident neutron energy. However, for most nuclei the imaginary part of the scattering length is small and the scattering length can be considered to be real. We shall here confine our discussion to such nuclei.

The scattering cross-section $d\sigma/d\Omega$ for scattering of neutrons from a single fixed nucleus can be readily calculated. If v is the velocity of the neutron which is the same before and after scattering (elastic), then the number of neutrons passing through the area dS per second is

$$v dS |\psi_{\text{sc}}|^2 = v dS \frac{b^2}{r^2} = vb^2 d\Omega. \quad (27)$$

The flux of incident neutrons is given by

$$\Phi = v |\psi_{\text{inc}}|^2 = v. \quad (28)$$

We therefore get from the definition of cross-section

$$\frac{d\sigma}{d\Omega} = \frac{vb^2 d\Omega}{\Phi d\Omega} = b^2. \quad (29)$$

The total scattering cross-section σ_{tot} is given by

$$\sigma_{\text{tot}} = 4\pi b^2. \quad (30)$$

The value of the neutron scattering length of a nucleus depends on the spin state of the nucleus–neutron system. For a nucleus with spin I the spin of the nucleus–neutron system is either $I + 1/2$ or $I - 1/2$. Each spin state has its own value of b , denoted by b^+ and b^- corresponding to the spin states $I + 1/2$ and $I - 1/2$, respectively. If the spin of the nucleus is zero, the nucleus–neutron system can only have spin $1/2$ and therefore there is only one value of the scattering length. There exists no sufficiently accurate theory of nuclear forces so far to allow scattering lengths to be calculated. Fortunately scattering lengths of most nuclei have been determined experimentally and have been tabulated in the literature [19]. The neutron scattering length depends on the details of the interaction between the neutron and the components of the nucleus and therefore the sign and the magnitude of b change in an irregular fashion with the atomic number. This irregularity in neutron scattering length is due to the superposition of resonance scattering with the slowly increasing potential scattering as a function of the atomic weight. This is in contrast to the X-ray scattering factor which increases monotonically with the atomic number. Figure 2 shows the values of bound-atom neutron scattering lengths of elements as a function of the atomic number. The values shown are for naturally occurring isotopic compositions. The irregular variation of the neutron scattering length has some important consequences:

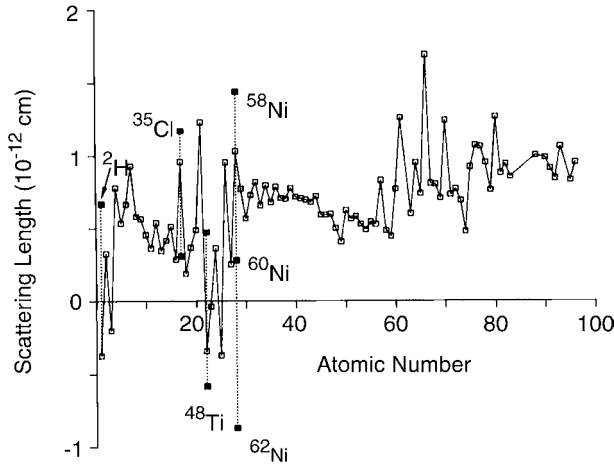


Fig. 2. Values of bound-atom neutron scattering lengths of elements as a function of the atomic number. The values shown are for the naturally occurring isotopic compositions. The solid squares indicate some values for separated isotopes often used in the isotope-substitution experiments (after Price and Sköld [15]).

1. Unlike the case of X-ray scattering, the neutron scattering lengths of light atoms, like hydrogen and oxygen, are quite large and therefore their positions can be easily determined in presence of heavier atoms (see Figures 2 and 3).
2. Neutrons can distinguish between atoms of comparable atomic number.
3. Neutrons can usually distinguish isotopes of the same element due to their different scattering lengths. These properties sometimes lead to the choice of the neutrons rather than X-rays as the favorable probe for structural investigations.

5.2. Coherent and incoherent scattering

Neutron scattering from naturally occurring elements, which are often composed of different isotopes with zero and nonzero nuclear spins I , is more complex than that discussed in the previous section. The scattering lengths depend not only on the kind of element, but also on the kind of isotope and the quantum number of the angular momentum $I \pm 1/2$ of nucleus–neutron system. The outgoing partial neutron waves from the individual nuclei attain different phases and amplitudes and do not interfere completely anymore. Hence one must distinguish between coherent and isotope and/or spin-incoherent processes. The scattering amplitude operator for a particular isotope α is given by

$$b = A_{\alpha} + B_{\alpha} \sigma \cdot I_{\alpha}, \quad (31)$$

where σ is the Pauli spin operator of the neutron and the constants A_{α} and B_{α} are isotope specific constants. The coherent scattering cross-section is given by the square of the

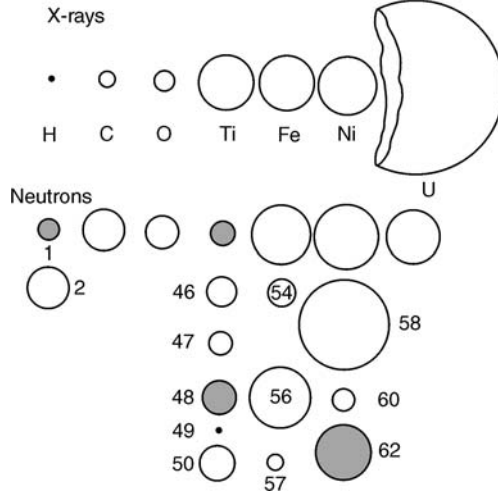


Fig. 3. Schematic representation of the scattering cross-sections of elements for X-rays and neutrons. The radii of the circles are proportional to the scattering amplitude b . Negative values of b are indicated by the cross-hatched shading (after Bacon [11]).

average of the scattering lengths

$$\sigma_c = 4\pi \left(\frac{1}{N} \sum_{i=1}^N b_i \right)^2 = 4\pi \langle b \rangle^2. \quad (32)$$

The coherent processes are, for example, Bragg scattering and inelastic scattering by phonons and magnons. The incoherent scattering cross-section arises, as we have already pointed out, due to the disorder of the scattering lengths of chemically identical particles and is given by

$$\sigma_i = 4\pi (\langle b^2 \rangle - \langle b \rangle^2). \quad (33)$$

Diffusions, crystal-field excitations, Stoner excitations and also nuclear excitations due to hyperfine field splitting are examples of such incoherent processes.

6. Magnetic neutron scattering

6.1. Scattering of neutrons from unpaired electrons

We first wish to derive an expression for the interaction potential due to the magnetic interaction between a neutron in spin state σ and a moving electron of momentum $\mathbf{p}$ and spin $\mathbf{s}$. The corresponding magnetic moment of the neutron is $\boldsymbol{\mu}_n = -\gamma \mu_N \boldsymbol{\sigma}$ and that of the electron is $\boldsymbol{\mu}_e = -2\mu_B \mathbf{s}$, where $\gamma = 1.9132$ and μ_N and μ_B are the nuclear magneton

and the Bohr magneton. The interaction potential is

$$\mathbf{V}(\mathbf{r}) = -\gamma\mu_N 2\mu_B \boldsymbol{\sigma} \cdot \left[\text{curl} \left(\frac{\mathbf{s} \times \hat{\mathbf{R}}}{R^2} \right) + \frac{1}{\hbar} \frac{\mathbf{p} \times \hat{\mathbf{R}}}{R^2} \right]. \quad (34)$$

The interaction potential consists of two terms: the first term is due to spin and the second due to the orbital motion of the electron.

To calculate the scattering cross-section (23) from a sample containing many unpaired electrons we must calculate the matrix elements $\langle \mathbf{k}_1 \sigma_1 \lambda_1 | V | \mathbf{k}_0 \sigma_0 \lambda_0 \rangle$. The matrix element $\langle \mathbf{k}_1 | V | \mathbf{k} \rangle$ is given by

$$\sum_i \langle \mathbf{k}_1 | V | \mathbf{k}_0 \rangle = 4\pi\gamma\mu_N 2\mu_B \boldsymbol{\sigma} \cdot \mathbf{D}_\perp (\mathbf{k}_0 - \mathbf{k}_1), \quad (35)$$

where

$$\mathbf{D}_\perp = \sum_i \left\{ \hat{\mathbf{Q}} \times (\mathbf{s}_i \times \hat{\mathbf{Q}}) + \frac{i}{\hbar Q} (\mathbf{p}_i \times \hat{\mathbf{Q}}) \right\} \exp(i\mathbf{Q} \cdot \mathbf{r}_i). \quad (36)$$

The scattering vector $\mathbf{Q}$ is given by

$$\mathbf{Q} = \mathbf{k}_1 - \mathbf{k}_0 \quad (37)$$

and $\hat{\mathbf{Q}}$ is the unit vector along $\mathbf{Q}$. The operator $\mathbf{D}_\perp$ is called the magnetic interaction operator which is related to the magnetization of the scattering system and can be written as

$$\mathbf{D}_\perp = \mathbf{D}_{\perp S} + \mathbf{D}_{\perp L} = \hat{\mathbf{Q}} \times (\mathbf{D} \times \hat{\mathbf{Q}}), \quad (38)$$

where

$$\mathbf{D} = \mathbf{D}_S + \mathbf{D}_L = -\frac{1}{2\mu_B} \mathbf{M}(\mathbf{Q}). \quad (39)$$

The subscripts “S” and “L” refer to the spin and the orbital parts. $\mathbf{M}(\mathbf{Q})$ can be written as

$$\mathbf{M}(\mathbf{Q}) = \mathbf{M}_S(\mathbf{Q}) + \mathbf{M}_L(\mathbf{Q}) = \int \mathbf{M}(\mathbf{r}) \exp(i\mathbf{Q} \cdot \mathbf{r}_i) d\mathbf{r}, \quad (40)$$

where

$$\mathbf{M}(\mathbf{r}) = \mathbf{M}_S(\mathbf{r}) + \mathbf{M}_L(\mathbf{r}) \quad (41)$$

is the total magnetization operator. The operator $\mathbf{D}$ is the Fourier transform of $\mathbf{M}(\mathbf{r})$ and $\mathbf{D}_\perp$ is the projection of $\mathbf{D}$ on to the plane perpendicular to $\hat{\mathbf{Q}}$. They are related by

$$\mathbf{D}_\perp = \mathbf{D} - (\mathbf{D} \cdot \hat{\mathbf{Q}}) \hat{\mathbf{Q}}. \quad (42)$$

We have therefore

$$\mathbf{D}_{\perp}^+ \cdot \mathbf{D}_{\perp} = \sum_{\alpha\beta} (\delta_{\alpha\beta} - \hat{\mathbf{Q}}_{\alpha} \hat{\mathbf{Q}}_{\beta}) \mathbf{D}_{\alpha}^+ \mathbf{D}_{\beta}, \quad (43)$$

where α and β stand for x, y, z and $\delta_{\alpha\beta}$ is the Kronecker delta.

It is now necessary to evaluate the matrix elements over the neutron spin states. For unpolarized neutrons the products of the neutron spin operator σ satisfy the equation

$$\sum_{\sigma_0} p_{\sigma_0} \sum_{\sigma_1} |\langle \sigma_1 | \sigma_{\alpha} \sigma_{\beta} | \sigma_0 \rangle|^2 = \delta_{\alpha\beta}. \quad (44)$$

Now from the master formula (23) we get

$$\frac{d^2\sigma}{d\Omega dE} = \frac{1}{N_m} \frac{k_1}{k_0} (\gamma r_0)^2 \sum_{\alpha} \sum_{\lambda_0} p_{\lambda_0} \sum_{\lambda_1} \langle \lambda_0 | \mathbf{D}_{\perp\alpha}^+ | \lambda_1 \rangle \langle \lambda_1 | \mathbf{D}_{\perp\alpha} | \lambda_0 \rangle \delta(E + E_{\lambda_0} - E_{\lambda_1}), \quad (45)$$

where $r_0 = e^2/(m_e c^2) = 0.28179 \times 10^{-12}$ cm is the classical radius of the electron and N_m is the number of magnetic ions. Introducing time dependence equation (45) can be rewritten as

$$\frac{d^2\sigma}{d\Omega dE} = \frac{1}{N_m} \frac{k_1}{k_0} \frac{(\gamma r_0)^2}{2\pi\hbar} \sum_{\alpha} \int \langle \mathbf{D}_{\perp\alpha}^+(-\mathbf{Q}, 0) \mathbf{D}_{\perp\alpha}(\mathbf{Q}, t) \rangle e^{-iEt/\hbar} dt. \quad (46)$$

Using (43) we can rewrite (46) as

$$\frac{d^2\sigma}{d\Omega dE} = \frac{1}{N_m} \frac{k_1}{k_0} \frac{(\gamma r_0)^2}{2\pi\hbar} \sum_{\alpha\beta} (\delta_{\alpha\beta} - \hat{\mathbf{Q}}_{\alpha} \hat{\mathbf{Q}}_{\beta}) \int \langle \mathbf{D}_{\alpha}(-\mathbf{Q}, 0) \mathbf{D}_{\beta}(\mathbf{Q}, t) \rangle e^{-iEt/\hbar} dt. \quad (47)$$

So the scattering is dependent on the magnetic fluctuation of the scattering system. This is analogous to the nuclear scattering which is dependent on the density fluctuation of the system. But what distinguishes the magnetic scattering is its directional dependence through the tensor $(\delta_{\alpha\beta} - \hat{\mathbf{Q}}_{\alpha} \hat{\mathbf{Q}}_{\beta})$ which picks out the components of the magnetization perpendicular to the momentum transfer $\mathbf{Q}$. If the direction of magnetization in the scattering system is unique and if we define a unit vector $\hat{\boldsymbol{\eta}}$ along this direction then the directional dependence of the magnetic scattering is given by the factor $1 - (\hat{\mathbf{Q}} \cdot \hat{\boldsymbol{\eta}})^2$. In fact, in favorable cases this factor enables us to determine the orientation of the magnetic moments in crystalline materials.

6.2. Scattering of neutrons from crystalline magnetic materials

So far we have made no assumption about the spatial distribution of the unpaired electrons and have restricted ourselves to the general case of a many-body system of unpaired electrons. Now we wish to consider the scattering of neutrons from crystalline magnetic systems. At first we will restrict ourselves to a localized electron system in which the unpaired electrons are situated on the magnetic ions which are arranged in the crystalline lattice. If the magnetic atoms are localized on the positions $\mathbf{R}_i$ and the electrons have $\mathbf{LS}$ coupling, then the time dependent magnetic interaction operator can be given by

$$\mathbf{D}(\mathbf{Q}, t) = \sum_i f_i(\mathbf{Q}) \boldsymbol{\mu}_i(t) e^{i\mathbf{Q} \cdot \mathbf{R}_i(t)}, \quad (48)$$

where $\boldsymbol{\mu}_i = \frac{1}{2} \mathbf{S}_i$ is the magnetic moment associated with the atom at $\mathbf{R}_i$ in units of μ_B and $f_i(\mathbf{Q})$ are the atomic form factors defined by

$$f(\mathbf{Q}) = \frac{g_S}{g} \bar{j}_0(\mathbf{Q}) + \frac{g_L}{g} [\bar{j}_0(\mathbf{Q}) + \bar{j}_2(\mathbf{Q})], \quad (49)$$

with

$$\bar{j}_n(\mathbf{Q}) = \int j_n(Qr) |\psi(r)|^2 d\mathbf{r}, \quad (50)$$

where g , g_L and g_S are the gyromagnetic ratios, j_n are spherical Bessel functions and ψ are the atomic wave functions. The $f_i(\mathbf{Q})$ are normalized such that $f_i(\mathbf{Q} = 0) = 1$. So the magnitude and the orientation of $\boldsymbol{\mu}_i(t)$ are the magnetic moment value in Bohr magnetons and its spatial orientation and $f_i(\mathbf{Q})$ is the Fourier transform of the magnetic electron density about the atomic center. Calculated values of j_n have been tabulated by Brown [20]. Now assuming that the interatomic forces are not affected by magnetic moments, one can carry out the sums over electronic and nuclear coordinates separately. Under this assumption the scattering cross-section can be written as

$$\begin{aligned} \frac{d^2\sigma}{d\Omega dE} &= \frac{1}{N_m} \frac{k_1}{k_0} \frac{(\gamma r_0)^2}{2\pi\hbar} \sum_{\alpha\beta} (\delta_{\alpha\beta} - \hat{\mathbf{Q}}_\alpha \hat{\mathbf{Q}}_\beta) \sum_{ii'} f_i^*(\mathbf{Q}) f_{i'}(\mathbf{Q}) \\ &\times \int \langle \mu_{i\alpha}(0) \mu_{i'\beta}(t) \rangle \langle e^{-i\mathbf{Q} \cdot \mathbf{R}_i(0)} e^{i\mathbf{Q} \cdot \mathbf{R}_{i'}(t)} \rangle e^{-iEt/\hbar} dt. \end{aligned} \quad (51)$$

The above expression still contains the correlation function for the atomic coordinates. Therefore the magnetic scattering is not only influenced by the magnetic behavior of the scattering systems but is also influenced by the crystalline structure and dynamics.

6.3. Elastic magnetic scattering from crystals

We have seen in the previous section that the magnetic scattering is dependent on the spin-spin correlation function. When the magnetic moments of the atoms order at low tempera-

ture then the spins are correlated. If these correlations persist for infinite time the scattering is purely elastic and correlations are time independent. In such case we get from (46)

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{el}} = \frac{1}{N_m}(\gamma r_0)^2 |\langle \mathbf{D}_\perp(\mathbf{Q}) \rangle|^2. \quad (52)$$

For a crystal we can put

$$\mathbf{R}_i(t) = \mathbf{R}_\ell + \mathbf{r}_d + \mathbf{u}_i(t), \quad (53)$$

where $\mathbf{R}_\ell$ is the coordinate of the ℓ th unit cell, $\mathbf{r}_d$ is the equilibrium position of the atom inside the unit cell and $\mathbf{u}_i(t)$ is the displacement of the atom from the equilibrium position. If $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are the direct unit cell vectors we have

$$\mathbf{R}_\ell = \ell_1 \mathbf{a} + \ell_2 \mathbf{b} + \ell_3 \mathbf{c} \quad (54)$$

and

$$\mathbf{r}_d = x_d \mathbf{a} + y_d \mathbf{b} + z_d \mathbf{c}. \quad (55)$$

The reciprocal lattice vector $\boldsymbol{\tau}$ is defined as

$$\boldsymbol{\tau} = h \mathbf{a}^* + k \mathbf{b}^* + l \mathbf{c}^*, \quad (56)$$

where $\mathbf{a}^*$, $\mathbf{b}^*$, $\mathbf{c}^*$ are the reciprocal unit cell vectors given by

$$\mathbf{a}^* = \frac{2\pi}{v_0}(\mathbf{b} \times \mathbf{c}), \quad \mathbf{b}^* = \frac{2\pi}{v_0}(\mathbf{c} \times \mathbf{a}), \quad \mathbf{c}^* = \frac{2\pi}{v_0}(\mathbf{a} \times \mathbf{b}). \quad (57)$$

The reciprocal lattice vector $\boldsymbol{\tau}$ is related to the reciprocal lattice vector $\mathbf{H}$ normally used by crystallographers by $\boldsymbol{\tau} = 2\pi\mathbf{H}$. In general the unit cell contains several kinds of crystallographically distinct atoms located in different sites (see International Tables for Crystallography [21]). So the index d must be split into $d = (j, s)$, where j is the index of atom in the site s . The number of sites is denoted by n_s and for a given site the order of the site n_B gives the number of Bravais lattices for that site.

We get from (48) after averaging nuclear and electronic parts independently

$$\langle \mathbf{D}(\mathbf{Q}) \rangle = \frac{1}{\gamma r_0} \sum_{\ell} \mathbf{F}_M(\mathbf{Q}) e^{i\mathbf{Q} \cdot \mathbf{R}_\ell}, \quad (58)$$

where

$$\mathbf{F}_M(\mathbf{Q}) = \gamma r_0 \sum_d f_d(\mathbf{Q}) \langle \boldsymbol{\mu}_d \rangle e^{i\mathbf{Q} \cdot \mathbf{r}_d} e^{-W_d(\mathbf{Q})} \quad (59)$$

is the magnetic unit cell structure factor. The factor $e^{-W_d(\mathbf{Q})}$ is the well-known Debye–Waller factor which arises due to the thermal motion of the atoms. The differential magnetic cross-section can be written as

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{el}} = \frac{1}{N_m} \frac{2\pi^3}{v_0} \sum_{\tau_M} \delta(\mathbf{Q} - \tau_M) |\mathbf{F}_{M\perp}(\tau_M)|^2, \quad (60)$$

where

$$\mathbf{F}_{M\perp} = \hat{\mathbf{Q}} \times \mathbf{F}_M \times \hat{\mathbf{Q}} \quad (61)$$

is the magnetic structure factor and τ_M is the reciprocal lattice vector of the magnetic structure which can be written as $\tau_M = \tau \pm \mathbf{k}$. The vector $\mathbf{k}$ characterizes the magnetic periodicity and is called the propagation or wave vector of the magnetic structure. Just as in the case of the nuclear Bragg reflection, the magnetic Bragg reflection arises for $\mathbf{Q} = \tau_M$.

6.4. Inelastic magnetic scattering

We now consider inelastic magnetic scattering of neutrons. Inelastic magnetic neutron scattering, like the inelastic nuclear scattering, can originate either from single particle excitations like crystal-field excitations or collective excitations like spin waves. The starting equation for inelastic magnetic scattering is equation (51) which we repeat here

$$\begin{aligned} \frac{d^2\sigma}{d\Omega dE} &= \frac{1}{N_m} \frac{k_1}{k_0} \frac{(\gamma r_0)^2}{2\pi\hbar} \sum_{\alpha\beta} (\delta_{\alpha\beta} - \hat{\mathbf{Q}}_\alpha \hat{\mathbf{Q}}_\beta) \sum_{ii'} f_i^*(\mathbf{Q}) f_{i'}(\mathbf{Q}) \\ &\times \int_{-\infty}^{\infty} \langle \mu_{i\alpha}(0) \mu_{i'\beta}(t) \rangle \langle e^{-i\mathbf{Q}\cdot\mathbf{R}_i(0)} e^{i\mathbf{Q}\cdot\mathbf{R}_{i'}(t)} \rangle e^{-iEt/\hbar} dt. \end{aligned} \quad (62)$$

We can split the two correlation functions into a time-independent part plus the part giving the time dependence at finite times:

$$\begin{aligned} \mathbf{J}_{ii'}^{\alpha\beta}(t) &= \langle \mu_{i\alpha}(0) \mu_{i'\beta}(t) \rangle = \mathbf{J}_{ii'}^{\alpha\beta}(\infty) + \mathbf{J}_{ii'}^{\prime\alpha\beta}(t), \\ \mathbf{I}_{ii'}(\mathbf{Q}, t) &= \langle e^{-i\mathbf{Q}\cdot\mathbf{R}_i(0)} e^{i\mathbf{Q}\cdot\mathbf{R}_{i'}(t)} \rangle = \mathbf{I}_{ii'}(\mathbf{Q}, \infty) + \mathbf{I}'_{ii'}(\mathbf{Q}, t). \end{aligned} \quad (63)$$

The integral in (62) can be written as

$$\int [\mathbf{J}_{ii'}^{\alpha\beta}(\infty) + \mathbf{J}_{ii'}^{\prime\alpha\beta}(t)] [\mathbf{I}_{ii'}(\mathbf{Q}, \infty) + \mathbf{I}'_{ii'}(\mathbf{Q}, t)] e^{-iEt/\hbar} dt. \quad (64)$$

The quantities are in the two square brackets when multiplied give four terms. The term $\mathbf{J}_{ii'}^{\alpha\beta}(\infty) \mathbf{I}_{ii'}(\mathbf{Q}, \infty)$ gives rise to elastic scattering and has already been discussed. The term $\mathbf{J}_{ii'}^{\alpha\beta}(\infty) \mathbf{I}'_{ii'}(\mathbf{Q}, t)$ is elastic in the spin system but inelastic in the phonon system and is

known as magnetovibrational scattering. Here the orientation of the spin system remains unchanged but the neutron excites or de-excites phonons via magnetic interaction. The term $\mathbf{J}_{ii'}^{\alpha\beta}(t)\mathbf{I}_{ii'}(\mathbf{Q}, \infty)$ is inelastic in the spin system with no change in the phonon system whereas $\mathbf{J}_{ii'}^{\alpha\beta}(t)\mathbf{I}_{ii'}'(\mathbf{Q}, t)$ gives scattering which is inelastic in both the spin and phonon systems. Here we are mainly concerned with the term $\mathbf{J}_{ii'}^{\alpha\beta}(t)\mathbf{I}_{ii'}(\mathbf{Q}, \infty)$ which gives inelastic magnetic scattering. For a crystalline solid with long range order the total (elastic plus inelastic) magnetic cross-section is given by

$$\frac{d^2\sigma}{d\Omega dE} = \frac{k_1}{k_0} \frac{(\gamma r_0)^2}{4\pi\mu_B^2} \sum_{\alpha\beta} (\delta_{\alpha\beta} - \hat{\mathbf{Q}}_\alpha \hat{\mathbf{Q}}_\beta) (1 - e^{-E/(k_B T)})^{-1} \times \sum_{dd'} e^{-[W_{d'}(\mathbf{Q}) + W_d(\mathbf{Q})]} (\chi''_{\alpha\beta})^{dd'}(\mathbf{Q}, E), \quad (65)$$

where χ'' is the imaginary part of the generalized susceptibility $\chi_{\alpha\beta}^{dd'}(\mathbf{Q}, E)$ defined by

$$\begin{aligned} (\chi'')_{\alpha\beta}^{dd'}(\mathbf{Q}, E) &= 4\pi\mu_B^2 f_d^*(\mathbf{Q}) f_{d'}(\mathbf{Q}) (1 - e^{-E/(k_B T)}) \\ &\times \sum_{\ell} e^{i\mathbf{Q}\cdot\mathbf{R}_\ell} e^{i\mathbf{Q}\cdot(\mathbf{r}_{d'} - \mathbf{r}_d)} \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \langle \mu_{0d\alpha}(0) \mu_{\ell d'\beta}(t) \rangle e^{-iEt/\hbar} dt. \end{aligned} \quad (66)$$

The factor $(1 - e^{-E/(k_B T)})$ in (66) maintains the principle of detailed balance. The generalized susceptibility $\chi''(\mathbf{Q}, E)$ plays a similar role to that of $S(\mathbf{Q}, E)$ for inelastic nuclear scattering; it contains the detailed information about the dynamics of the spin system.

6.5. Spin waves

Analogous to the normal modes of atomic displacements or phonons, spin deviations in a magnetic system may be represented by the sum of deviations due to the set of traveling sinusoidal waves. These are the spin waves. The concept of spin waves can be illustrated by Figure 4. The classical picture of the ground state of a simple ferromagnet has all spins parallel as shown in Figure 4(a). Consider N spins each of S on a line or a ring with nearest neighbors separated by a distance a and coupled by the Heisenberg interaction given by

$$H = -2J \sum_i \mathbf{S}_i \cdot \mathbf{S}_{i+1}. \quad (67)$$

A possible excitation of the ferromagnet has been illustrated in Figure 4(b) in which an isolated spin is reversed. This is an unlikely situation because it costs an energy $8JS^2$ which is considerable. A low-lying excitation which costs much less energy can be formed if we let all spins share the reversal as has been illustrated in Figure 4(c). In a classical

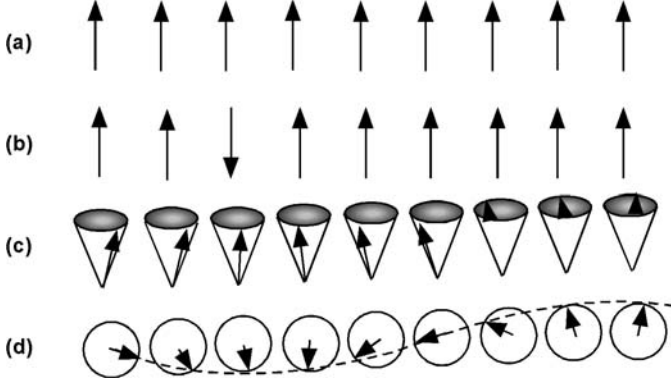


Fig. 4. (a) Ground state spin alignment of a simple ferromagnet. (b) An isolated spin is reversed which costs a lot of energy. (c) A lower energy is achieved by sharing the disturbance among neighboring atoms. (d) A spin wave that describes their spin direction is clearly seen in the plan view (after Bacon [11]).

picture the spin vectors precess on the surface of cones, with each successive spin advanced in phase by a constant angle such that the variation of the spin vectors forms a wave. Figure 4(d) shows the spins as viewed from above at a particular instant. The excitation illustrated in Figures 4(c) and 4(d) has a wavelike form and is called a spin wave or when quantized, a magnon. There will be a series of such waves and each spin wave will possess a wavelength and a quantized energy $\hbar\omega$. This corresponds to the normal modes of a coupled spin system whose energy $E = \hbar\omega$ and the wave vector q are given by the dispersion relation. For a ferromagnetic linear chain with nearest neighbor interaction this relation is given by

$$\hbar\omega = 4JS(1 - \cos qa). \quad (68)$$

At long wavelength (low energy) $qa \ll 1$ so that $(1 - \cos qa) \simeq \frac{1}{2}(qa)^2$. In this limit the dispersion relation for the spin wave becomes

$$\hbar\omega \simeq 2JSa^2q^2 = Dq^2, \quad (69)$$

where D is called the spin wave stiffness constant. Figure 5(a) shows the typical spin wave dispersion of a ferromagnet. For an antiferromagnetic chain the corresponding dispersion relation is given by

$$\hbar\omega = 4|J|S \sin qa. \quad (70)$$

For the long-wavelength limit this equation is reduced to

$$\hbar\omega \simeq 4|J|Sa q. \quad (71)$$

We note the difference between the dispersion of spin waves for a ferromagnet and an antiferromagnet. For small q the spin wave energy $\hbar\omega$ is proportional to q^2 for a ferromagnet

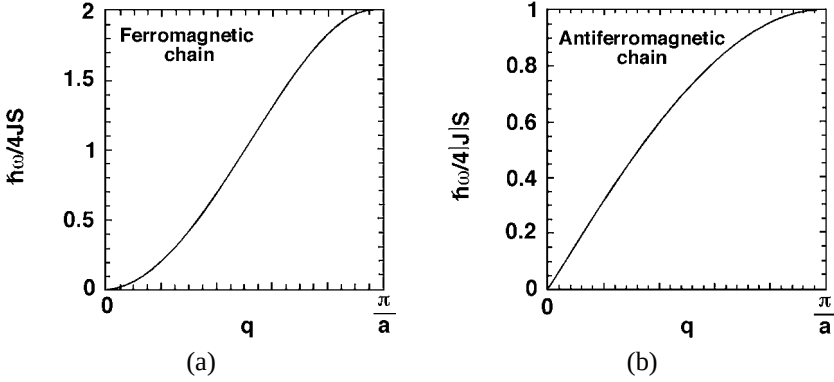


Fig. 5. (a) Spin wave dispersion of a ferromagnetic chain. (b) Spin wave dispersion of an antiferromagnetic chain.

and is proportional to q for an antiferromagnet. Figure 5(b) shows the typical dispersion curve of an antiferromagnet. In Figures 5(a) and 5(b) there are no gaps in the dispersion so that the energy $\hbar\omega$ is zero at $q = 0$. In many cases there exist such gaps caused by the single-ion anisotropy or the anisotropy in the exchange interaction.

We have given above the simple examples of one-dimensional linear ferromagnetic and antiferromagnetic chains. For a more general case the dispersion of the spin waves can be written as

$$\hbar\omega(q) = 2S[\mathcal{J}(0) - \mathcal{J}(q)] \quad (72)$$

in terms of the Fourier-transformed exchange coupling

$$\mathcal{J}(q) = \sum_{\ell'} J_{\ell\ell'} e^{iq \cdot (\mathbf{R}_\ell - \mathbf{R}_{\ell'})}. \quad (73)$$

6.6. Scattering from spin waves

For a Bravais lattice the generalized susceptibility can be written as

$$\begin{aligned} \chi''_{\alpha\beta}(\mathbf{Q}, E) &= \pi g^2 \mu_B^2 (1 - e^{-E/(k_B T)}) \\ &\times \sum_{\ell} e^{i\mathbf{Q} \cdot \mathbf{R}_\ell} \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \langle S_{0\alpha}(0) S_{\ell\beta}(t) \rangle e^{-iEt/\hbar} dt. \end{aligned} \quad (74)$$

It can be shown that the only nonzero components of $\langle S_{0\alpha}(0) S_{\ell\beta}(t) \rangle$ are

$$\begin{aligned} \langle S_{0x}(0) S_{\ell x}(t) \rangle &= \langle S_{0y}(0) S_{\ell y}(t) \rangle \\ &= \frac{S}{2N} \sum_{\mathbf{q}} [e^{-i[\mathbf{q} \cdot \mathbf{R}_\ell - \omega(\mathbf{q})t]} \langle n_{\mathbf{q}} + 1 \rangle + e^{i[\mathbf{q} \cdot \mathbf{R}_\ell - \omega(\mathbf{q})t]} \langle n_{\mathbf{q}} \rangle], \end{aligned} \quad (75)$$

$$\begin{aligned}\langle S_{0x}(0)S_{\ell y}(t) \rangle &= -\langle S_{0y}(0)S_{\ell x}(t) \rangle \\ &= \frac{iS}{2N} \sum_{\mathbf{q}} [e^{-i\{\mathbf{q}\cdot\mathbf{R}_{\ell}-\omega(\mathbf{q})t\}} \langle n_{\mathbf{q}} + 1 \rangle - e^{i\{\mathbf{q}\cdot\mathbf{R}_{\ell}-\omega(\mathbf{q})t\}} \langle n_{\mathbf{q}} \rangle],\end{aligned}\quad (76)$$

$$\langle S_{0z}(0)S_{\ell z}(t) \rangle = S^2 - \frac{2S}{N} \sum_{\mathbf{q}} \langle n_{\mathbf{q}} \rangle, \quad (77)$$

where

$$\langle n_{\mathbf{q}} \rangle = \frac{1}{\exp[\hbar\omega(\mathbf{q})/(k_B T)] - 1} \quad (78)$$

is the population factor. The (z, z) term is time independent which gives elastic scattering. The (x, y) and (y, x) terms cancel in the expression for the neutron scattering cross-section given in (65). The inelastic neutron scattering cross-section from the spin waves is given by the sum of terms for one magnon creation and annihilation:

$$\begin{aligned}\left(\frac{d^2\sigma}{d\Omega dE} \right)_{m,+1} &= \frac{k_1}{k_0} (\gamma r_0)^2 \frac{(2\pi)^3}{2N_m v_0} \frac{g^2 S}{4} (1 + \widehat{\mathbf{Q}}_z^2) f^2(\mathbf{Q}) e^{-2W(\mathbf{Q})} \\ &\quad \times \sum_{\tau\mathbf{q}} \langle n_{\mathbf{q}} + 1 \rangle \delta(\mathbf{Q} - \mathbf{q} - \boldsymbol{\tau}) \delta[E - \hbar\omega(\mathbf{q})],\end{aligned}\quad (79)$$

$$\begin{aligned}\left(\frac{d^2\sigma}{d\Omega dE} \right)_{m,-1} &= \frac{k_1}{k_0} (\gamma r_0)^2 \frac{(2\pi)^3}{2N_m v_0} \frac{g^2 S}{4} (1 + \widehat{\mathbf{Q}}_z^2) f^2(\mathbf{Q}) e^{-2W(\mathbf{Q})} \\ &\quad \times \sum_{\tau\mathbf{q}} \langle n_{\mathbf{q}} \rangle \delta(\mathbf{Q} + \mathbf{q} - \boldsymbol{\tau}) \delta[E + \hbar\omega(\mathbf{q})].\end{aligned}\quad (80)$$

From the above two equations we see that peaks in the inelastic neutron scattering from spin waves satisfy

$$\begin{aligned}\mathbf{Q} &= \boldsymbol{\tau} \pm \mathbf{q}, \\ E &= \pm \hbar\omega(\mathbf{q}).\end{aligned}\quad (81)$$

6.7. Paramagnetic scattering

In the true paramagnetic state of a localized magnetic system the magnetic moments of the ions are completely uncorrelated and are randomly oriented. This situation is perhaps never realized in practice but may be approximately true at temperatures much higher than the ordering temperature. For such a system

$$\langle \mu_{i\alpha}(0)\mu_{i'\beta}(t) \rangle = \langle \mu_{i\alpha}(0)^2 \rangle \delta_{ii'} \delta_{\alpha\beta} = \frac{1}{12} g_i^2 S_i(S_i + 1) \delta_{ii'} \delta_{\alpha\beta} \quad (82)$$

also $\sum_{\alpha\beta}(\delta_{\alpha\beta} - \hat{\mathbf{Q}}_\alpha \hat{\mathbf{Q}}_\beta) = 2$. We have from (51)

$$\begin{aligned} \frac{d^2\sigma}{d\Omega dE} &= \frac{1}{N_m} \frac{k_1}{k_0} \frac{(\gamma r_0)^2}{2\pi\hbar} \sum_i |f_i(\mathbf{Q})|^2 \frac{1}{6} g_i^2 S_i(S_i + 1) \\ &\times \int \langle e^{-i\mathbf{Q}\cdot\mathbf{R}_i(0)} e^{i\mathbf{Q}\cdot\mathbf{R}_i(t)} \rangle e^{-iEt/\hbar} dt. \end{aligned} \quad (83)$$

The scattering is similar to the nuclear incoherent scattering with the cross-section σ_{inc} replaced by $4\pi(\gamma r_0)^2 |f_i(\mathbf{Q})|^2 \frac{1}{6} g_i^2 S_i(S_i + 1)$. However, because of the Q dependence of the magnetic form factor the paramagnetic scattering has the same Q dependence, whereas the nuclear incoherent scattering has no such Q dependence. Due to the dipolar interaction between the neutron and the unpaired electronic spin, paramagnetic scattering can be separated from nuclear incoherent scattering by polarization analysis also.

If a magnetic field B is applied along the z direction, a nonzero average spin $\bar{S}_z$ leads to coherent scattering with an equivalent scattering length of $\gamma r_0 f_i(\mathbf{Q}) \frac{1}{2} g_i \bar{S}_z$. The coherent scattering will be superimposed on the nuclear Bragg scattering. The rest of the scattering will be just like paramagnetic scattering without a magnetic field but reduced in intensity.

For a perfect paramagnet, in which there exist no interactions and no correlations between the atomic spins, the magnetic scattering is spatially isotropic, elastic and is independent of temperature. However, the scattering from exchange-coupled paramagnets is quite different, even in the very high temperature limit for which there exist no short range correlations among the spins. For such an exchange-coupled system the neutron scattering cross-section is directly proportional to the generalized wave-vector-dependent susceptibility tensor and thus measures the Fourier components of the microscopic magnetization fluctuation. At a second-order phase transition a strong increase in the susceptibility is expected at the wave vector that characterizes the ultimate ordered phase. As the temperature of the system is decreased to the critical temperature from the high temperature side the inelasticity of the scattering decreases drastically. The magnetic structure of the ordered phase and therefore the propagation vector depends on the sign and strength of the exchange interactions and also the crystal structure. For a ferromagnet the ordering wave vector is $\mathbf{k} = 0$ and therefore corresponds to the nuclear Bragg positions. For a simple antiferromagnet the ordering vector is, for example (type-II cubic antiferromagnet), $\mathbf{k} = \langle \frac{1}{2} \frac{1}{2} \frac{1}{2} \rangle$ which corresponds to Bragg reflections for a lattice whose cell dimensions are twice those of the nuclear cell. Therefore the paramagnetic responses of ferro- and antiferromagnetically coupled systems are very different as one approaches the critical temperature.

6.8. Crystal-field excitations

Spin wave excitations are collective excitations in the magnetic system. Now we consider systems in which single ion effects, rather than the collective excitations, are important. The electrostatic and spin-orbit interactions lift the degeneracy of the unfilled $4f^n$ configuration of, for example, the rare-earth ions and give rise to the J -multiplets. Inelastic neutron scattering is a very powerful technique to determine the positions of energy levels and the matrix elements of the transitions between them. In inelastic neutron scattering

experiments both energy gain and the energy loss processes can be probed. The neutron can excite the rare-earth ion from a lower state to a higher state with a corresponding loss of the neutron energy or the rare-earth ion is de-excited from a higher energy level to a lower energy level and the neutron gains the corresponding energy. So the measured energy spectrum exhibits resonance peaks which can be associated with the transitions between the crystal-field-split levels. A unique identification of the observed energy levels is only possible by comparing the observed intensities of the resonance peaks with the expected neutron scattering cross-sections of different transitions among the levels. The neutron scattering cross-sections of such processes can be derived from (42), (45) and (48),

$$\begin{aligned} \frac{d^2\sigma}{d\Omega dE} &= \frac{1}{N_m} \frac{k_1}{k_0} (\gamma r_0)^2 \sum_{\alpha\beta} (\delta_{\alpha\beta} - \hat{\mathbf{Q}}_\alpha \hat{\mathbf{Q}}_\beta) |f(\mathbf{Q})|^2 e^{-2W} \\ &\times \frac{1}{Z} \sum_{\lambda_0 \lambda_1} e^{E_{\lambda_0}/(k_B T)} \langle \lambda_0 | \mu_\alpha^+ | \lambda_1 \rangle \langle \lambda_1 | \mu_\beta | \lambda_0 \rangle \delta(E + E_{\lambda_0} - E_{\lambda_1}), \quad (84) \end{aligned}$$

where Z is the partition function $\sum_{\lambda_0} e^{E_{\lambda_0}/(k_B T)}$. The inelastic neutron scattering investigations of the crystal field levels is important in cases where optical data are not available, for example, for metallic compounds. In neutron scattering investigations one is confronted with the problem of distinguishing the crystal-field excitation peaks from, say, phonon peaks. In such cases it is useful to check the temperature and Q dependence of the intensity of the inelastic peak. For phonon peaks the intensity is usually proportional to Q^2 whereas the intensity of the crystal field excitations decreases with Q according to $|f(\mathbf{Q})|^2$. The temperature dependence of the intensity of the inelastic peak is also useful for such purpose: phonons obey Bose statistics whereas the population of the crystal-field levels is governed by Boltzmann statistics. Polarized neutron scattering method is the surest technique for distinguishing phonon scattering from scattering due to the crystal field excitations. So far we have discussed crystal-field excitations as single particle excitations which have no dispersion. However, introduction of the interionic exchange interactions results in collective excitations with dispersion. These collective excitations are linear combinations of single-ion transitions, called magnetic excitons.

7. Concluding remarks

We have in this introductory chapter given only a very short account of the basic principles of magnetic neutron scattering and its potential applications. Some rudimentary account of the principles of nuclear neutron scattering has also been outlined. We have not derived the equations but merely listed the important ones. The following chapters will consider in more details many of the points, which are only touched upon here.

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